OS

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1. PRIORITY CPU SCHEDULING

dhanu@dhnpc: ~

GNU nano 7.2

```c
#include <stdio.h>

int main()
{
    int n, i, j;
    int bt[10], priority[10], p[10];
    int wt[10], tat[10];
    int temp;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time and priority:\n");

    for(i = 0; i < n; i++)
    {
        p[i] = i + 1;

        printf("P%d Burst Time: ", i + 1);
        scanf("%d", &bt[i]);

        printf("P%d Priority: ", i + 1);
        scanf("%d", &priority[i]);
    }

    /* Sort according to priority */
    for(i = 0; i < n - 1; i++)
    {
        for(j = i + 1; j < n; j++)
        {
            if(priority[i] > priority[j])
            {
                temp = priority[i];
                priority[i] = priority[j];
                priority[j] = temp;

                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;

                temp = p[i];
                p[i] = p[j];
                p[j] = temp;
            }
        }
    }

    wt[0] = 0;

    for(i = 1; i < n; i++)
    {
        wt[i] = wt[i - 1] + bt[i - 1];
    }

    for(i = 0; i < n; i++)
    {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tPriority\tWT\tTAT\n");

    for(i = 0; i < n; i++)
    {
        printf("P%d\t%d\t%d\t\t%d\t%d\n",
               p[i], bt[i], priority[i], wt[i], tat[i]);
    }

    return 0;
}
```

```
dhanu@dhnpc: ~
dhanu@dhnpc:~$ nano priority.c
dhanu@dhnpc:~$ nano priority.c
dhanu@dhnpc:~$ nano priority.c
gcc priority.c -o priority
./priority
Enter number of processes: 4
Enter burst time and priority:
P1 Burst Time: 1
P1 Priority: 1
P2 Burst Time: 2
P2 Priority: 3
P3 Burst Time: 1
P3 Priority: 2
P4 Burst Time: 3
P4 Priority: 2

Process BT      Priority        WT      TAT
P1      1       1               0       1
P3      1       2               1       2
P4      3       2               2       5
P2      2       3               5       7
dhanu@dhnpc:~$ |
```

2. ROUND ROBIN CPU SCHEDULING

```
dhanu@dhnpc: ~
  GNU nano 7.2
#include <stdio.h>
int main(){
    int n, i;
    int bt[10], rem[10];
    int tq, time = 0;
    int done;
  printf("Enter number of processes: ");
    scanf("%d", &n);
    printf("Enter burst time:\n");
    for(i = 0; i < n; i++)
    {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
        rem[i] = bt[i];
    }
    printf("Enter time quantum: ");
    scanf("%d", &tq);
    printf("\nExecution:\n");
    while(1)
    {
        done = 1;
        for(i = 0; i < n; i++)
        {
            if(rem[i] > 0)
            {
                done = 0;
                printf("P%d ", i + 1);
                if(rem[i] > tq)
                {
                    time = time + tq;
                    rem[i] = rem[i] - tq;
                }
                else
                {
                    time = time + rem[i];
                    rem[i] = 0; } } }
        if(done == 1)
            break;
    }

    printf("\nTotal time = %d\n", time);

    return 0;
}
```

```
dhanu@dhnpc: ~
dhanu@dhnpc:~$ nano roundrobin.c
dhanu@dhnpc:~$ gcc roundrobin.c -o roundrobin
dhanu@dhnpc:~$ ./roundrobin
Enter number of processes: 2
Enter burst time:
P1: 2
P2: 3
Enter time quantum: 2

Execution:
P1 P2 P2
Total time = 5
dhanu@dhnpc:~$
```

3. PROGRAM 3 — IPC using Pipe

```
dhanu@dhnpc: ~
  GNU nano 7.2

#include <stdio.h>
#include <unistd.h>
#include <string.h>

int main()
{
    int fd[2];
    char write_msg[] = "Hello from Parent";
    char read_msg[100];

    pipe(fd);

    if(fork() == 0)
    {
        close(fd[1]);

        read(fd[0], read_msg, sizeof(read_msg));

        printf("Child received: %s\n", read_msg);

        close(fd[0]);
    }
    else
    {
        close(fd[0]);

        write(fd[1], write_msg, strlen(write_msg) + 1);

        close(fd[1]);
    }

    return 0;
}
```

```
dhanu@dhnpc: ~
dhanu@dhnpc:~$ nano ipc.c
dhanu@dhnpc:~$ gcc ipc.c -o ipc
dhanu@dhnpc:~$ .\ipc
Command '.ipc' not found, did you mean:
  command 'tipc' from deb iproute2 (6.1.0-1ubuntu6.4)
Try: sudo apt install <deb name>
dhanu@dhnpc:~$ ./ipc
Child received: Hello from Parent
dhanu@dhnpc:~$
```

4. PROGRAM 4 — IPC using Shared Memory

```
dhanu@dhnpc: ~
  GNU nano 7.2

#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <string.h>

int main()
{
    int shmid;
    char *str;

    shmid = shmget(1234, 1024, 0666 | IPC_CREAT);

    str = (char *)shmat(shmid, NULL, 0);

    printf("Enter message: ");
    fgets(str, 100, stdin);

    printf("Message stored in shared memory: %s", str);

    shmdt(str);

    shmctl(shmid, IPC_RMID, NULL);

    return 0;
}
```

dhanu@dhnpc: ~

```
dhanu@dhnpc:~$ nano sharedmemory.c
dhanu@dhnpc:~$ gcc sharedmemory.c -o sharedmemory
dhanu@dhnpc:~$ .\sharedmemory
.sharedmemory: command not found
dhanu@dhnpc:~$ ./sharedmemory
Enter message: hello world
Message stored in shared memory: hello world
dhanu@dhnpc:~$
```